

The same participant with high miss-rates also had the poorest accuracy (87%). After excluding this participant, the filtered accuracy data were analyzed in a repeated-measures ANOVA, with attended hemifield (Left vs. Right) and Grouping (Grouped vs. Ungrouped stimuli) as repeated-measures factors. There was a significant main effect only for Attention, with higher accuracy for discriminating check-size changes in the left hemifield [$F_{(1, 7)} = 9.75, p = 0.02$]. Mean (and SD) for left was 94% (0.3%) and for right, 91% (1%). There was no significant interaction with grouping. A similar pattern was observed with the full data set.

WHOLE BRAIN ANALYSES

Statistical contrasts of left vs. right cued attention revealed significant activations (family-wise corrected for multiple comparisons at $p < 0.05$) contralateral to the cued hemifield in cuneus and lingual gyrus (see **Table 1** for coordinates). Lowered thresholds ($p < 0.001$ uncorrected, see **Figure 3**, left) revealed widespread activations only in posterior visual areas contralateral to the cued hemifield. Event-related contrasts of left vs. right transients showed significant activations contralateral to the transient (corrected for multiple comparisons at $p < 0.05$) in lingual gyrus and fusiform gyrus (**Table 1**). At lower thresholds (e.g., $p < 0.001$ uncorrected, **Figure 3**, right) activations were seen in occipital inferior and superior occipital areas putatively within area V3a (Tootell et al., 1995). Contrasts of grouped vs. ungrouped stimuli revealed no notable activations even at lowered thresholds ($p < 0.001$ uncorrected), for either event-related or block-related analyses. There were also no significant results for whole-brain block and event-related analyses of specific interactions between grouping and attention.

RETINOTOPY: BLOCKED ANALYSES

Beta weights (representing the overall level of BOLD activation) for each of the block-related conditions were estimated from each of the ROI's for each participant (i.e., masked retinotopic visual areas including hMT/V5+, see Methods). As we had no specific hypotheses about the laterality of attentional effects, data from left and right hemispheres were pooled according to whether the respective contralateral visual hemifield was attended or ignored. Data were entered into a 3-way repeated measures ANOVA with the following factors: Attention (whether a given ROI was contralateral vs. ipsilateral to the attended hemifield), Grouping, and cortical Area.

Table 1 | MNI coordinates (mm) of regions identified in whole-brain contrasts, significant at $p < 0.05$ corrected.

Contrast	Area	x^1	y	z	T
BLOCK-RELATED					
L-R	Cuneus	24	-94	20	6.72
R-L	Cuneus	-26	-92	29	8.22
	Lingual gyrus	-26	-80	-10	5.19
EVENT-RELATED					
L-R	Fusiform gyrus	32	-64	-18	7.12
R-L	Lingual gyrus	-26	-84	-12	4.31

¹ MNI coordinates (mm).

Initial analysis revealed a main effect of Attention, with larger BOLD signals when attention was cued to the contralateral vs. ipsilateral hemifield [$F_{(1, 7)} = 70.69, p = 0.0001$]; and a main effect of Grouping, with greater signals for ungrouped than grouped stimuli [$F_{(1, 6)} = 9.29, p = 0.019$]. There was a main effect of Area [$F_{(5, 35)} = 23.86, p < 0.0001$], which was partially accounted for by higher signal estimates in hMT/V5+ (9.87, SD 0.14) compared to the other areas (mean 4.86, SD 0.87). Variability was also much higher in MT/V5+ (SD 0.14) compared to other areas (mean SD 0.022, SD 0.017). The only significant interaction was between Attention and Area [$F_{(5, 35)} = 4.11, p = 0.005$], with larger effects of attention to the contralateral vs. ipsilateral hemifield in hMT/V5+ than in the other visual areas (see **Supplementary Figure 1**).

All further analyses excluded the participant with poor hit rates, in case this was indicative of poorly controlled attention. To render the variances between all ROI areas more uniform, we normalized each participant's beta estimates for each given condition from each given ROI (pooling data across hemispheres, as described above), by subtracting the average (and dividing by the standard deviation) of block-related beta estimates obtained across all conditions from the same bilateral ROIs. Normalized data now had similar means and ranges across ROIs and participants, while the detailed pattern of results across conditions remained unchanged within each ROI. An ANOVA based on the normalized data now revealed a significant interaction of Attention and Grouping [$F_{(1, 6)} = 8.06, p = 0.03$], with greater attentional modulation in the grouped compared to ungrouped condition (see **Figure 4**). These analyses also confirmed the main effect of Attention [$F_{(1, 6)} = 705, p < 0.0001$] and Grouping [$F_{(1, 6)} = 22.12, p = 0.003$], and no main effect or interaction for Area.

Separate analyses were conducted for each visual area showed that the interaction between grouping and attention was significant in V2 [$F_{(1, 6)} = 13.62, p = 0.01$], and borderline significant in V1 [$F_{(1, 6)} = 5.40, p = 0.059$] (see **Figure 5**). A

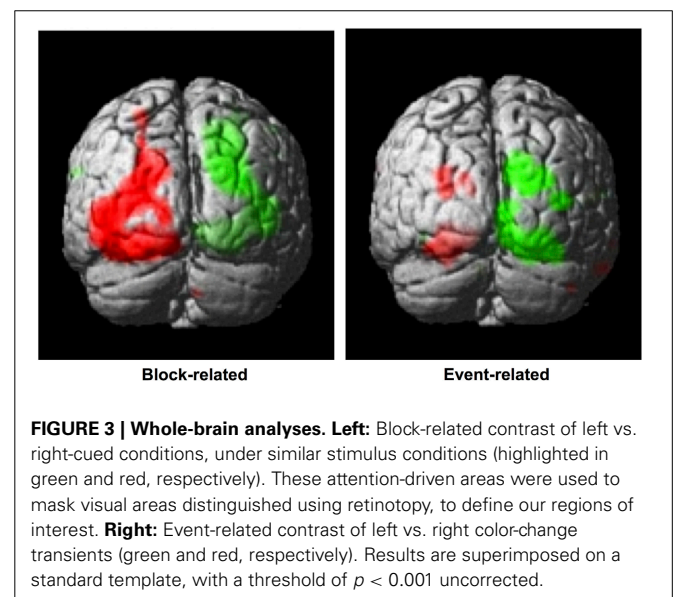


FIGURE 3 | Whole-brain analyses. Left: Block-related contrast of left vs. right-cued conditions, under similar stimulus conditions (highlighted in green and red, respectively). These attention-driven areas were used to mask visual areas distinguished using retinotopy, to define our regions of interest. **Right:** Event-related contrast of left vs. right color-change transients (green and red, respectively). Results are superimposed on a standard template, with a threshold of $p < 0.001$ uncorrected.